Theynujan Thangavelu

Hydrographic & Land Surveying | FIG/IHO/ICA Category B

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Summary

B.Sc. (Hons) Surveying Sciences graduate specialising in **Hydrographic Surveying**, from a programme recognised by IBSC at FIG/IHO/ICA Category B level. Field experience in **single-beam echosounder** and **side-scan sonar** surveys, plus land and topographic surveying with total station, GNSS/RTK and levels. Data processing in **HYPACK**, **Civil 3D**, **ArcGIS Pro** and **QGIS**. Currently a Hydrographic and Land Surveying Intern at NARA. Final-year research: **YOLOv11** framework for side-scan sonar object detection of subsea pipelines and communication cables.

Experience

National Aquatic Resources Research and Development Agency, Sri Lanka — July 2026 – Present

Hydrographic and Land Surveying Intern

- Surveyed and mapped existing sea cucumber farms at Gurunagar, Jaffna using GNSS/RTK and a shore-based total station tracking a prism on a survey boat to fix positions and spot heights in shallow water.
- Processed survey data in HYPACK to produce farm boundary maps.
- Conducted beach profile surveys at Erukkalampiddy, Mannar and Mulankavil, Kilinochchi, measuring cross-shore transects using GNSS RTK to record beach elevation changes for coastal monitoring and bathymetric chart preparation.

Sabaragamuwa University of Sri Lanka — 2023 – 2026

Field Surveying (Academic Fieldwork)

- Conducted single-beam echosounder bathymetric surveys at Gandara Fishery Harbour, including sound velocity profiling and processing in HYPACK.
- Carried out side-scan sonar survey at Gandara Fishery Harbour.
- Performed topographic and control surveys using total station, GNSS/RTK and levels.
- Conducted UAV (drone) survey of Gandara Fishery Harbour, Matara, processing imagery in Bentley ContextCapture to produce orthomosaic / 3D model / DSM.

Technical Skills

Survey Equipment: Single-Beam Echosounder (SBES), Side-Scan Sonar, Total Station, GNSS/RTK, UAV (Drone)

Hydrographic Software: HYPACK, MB System

Land Survey & CAD: Civil 3D, Leica Geo Office

GIS & Mapping: ArcGIS Pro, QGIS

Photogrammetry: Bentley ContextCapture

Programming: Python, Java, Dart/Flutter

Education & Certifications

Sabaragamuwa University of Sri Lanka — 2023 – 2026

B.Sc. (Hons) in Surveying Sciences (Hydrographic Surveying)

Colombo School of Computing — 2025 - Present

Bachelor of Information Technology (BIT)

Explore LiDAR in 3D Sep 2026
Coursera

ArcGIS Pro [2026]: Mapping, Spatial Analysis, 3D & Automation Aug 2026
Udemy (Dr. Abhijit Patil)

Python for Data Science, AI & Development Sep 2025
IBM (Coursera)

100 Days of Code: The Complete Python Pro Bootcamp Sep 2025
Udemy (Dr. Angela Yu)

References

Mr. A.N.D. Perera
Senior Lecturer, Faculty of Geomatics
Sabaragamuwa University of Sri Lanka
Hydrographic Surveying CAT A (FIG/IHO/ICA)

Mr. Roshan Ranaweera
Chief Hydrographer, Hydrographic Office
National Aquatic Resources Research and Development Agency